

Exercise 2: Tools, Agents and LCEL

Results

1. Terminal Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

• (langchain_project) user@LF-00002116:~/langchain_pro
/home/user/langchain_project/.venv/lib/python3.12/sit
`allowed_objects` will change in a future version. P
from langgraph.checkpoint.serde.jsonplus import Jsc
Extracted: The Coca-Cola Company
-> Inserted: The Coca-Cola Company
Extracted: Sony Corporation
-> Inserted: Sony Corporation
Extracted: McDonald's Corporation
-> Inserted: McDonald's Corporation
Extracted: Intel Corporation
-> Inserted: Intel Corporation
Extracted: Samsung Electronics Co., Ltd.
-> Inserted: Samsung Electronics Co., Ltd.
Extracted: Microsoft Corporation
-> Inserted: Microsoft Corporation
Extracted: Apple Inc.
-> Inserted: Apple Inc.
Extracted: Oracle Corporation
-> Inserted: Oracle Corporation
Extracted: NVIDIA Corporation
-> Inserted: NVIDIA Corporation
Extracted: Amazon.com Inc.
-> Inserted: Amazon.com Inc.
Extracted: Google LLC
-> Inserted: Google LLC
Extracted: Alibaba Group Holding Limited
-> Inserted: Alibaba Group Holding Limited
Extracted: SAP SE
-> Inserted: SAP SE
Extracted: LinkedIn Corporation
-> Inserted: LinkedIn Corporation
Extracted: Facebook, Inc.
-> Inserted: Facebook, Inc.
Extracted: Twitter, Inc.
-> Inserted: Twitter, Inc.
Extracted: Spotify AB
-> Inserted: Spotify AB
Extracted: YouTube LLC
-> Inserted: YouTube LLC
```

```
Extracted: Tesla, Inc.  
-> Inserted: Tesla, Inc.  
Extracted: Airbnb, Inc.  
-> Inserted: Airbnb, Inc.  
Extracted: PayPal Holdings, Inc.  
-> Inserted: PayPal Holdings, Inc.  
Extracted: Stripe, Inc.  
-> Inserted: Stripe, Inc.  
Extracted: Block, Inc.  
-> Inserted: Block, Inc.  
Extracted: Zoom Video Communications, Inc.  
-> Inserted: Zoom Video Communications, Inc.  
Extracted: Slack Technologies, LLC  
-> Inserted: Slack Technologies, LLC  
Extracted: Rivian Automotive, Inc.  
-> Inserted: Rivian Automotive, Inc.  
Extracted: SpaceX  
-> Inserted: SpaceX  
Extracted: TikTok  
-> Inserted: TikTok  
Extracted: ByteDance  
-> Inserted: ByteDance
```

```
****All companies processed!****
```

```
Total Companies: 29
```

```
○ (langchain_project) user@LF-00002116:~/langchain_project$
```

2. Database Output

```
postgres=# \c langchain_db
You are now connected to database "langchain_db" as user "postgres".
langchain_db=# SELECT * FROM company_details;
```

id	company_name	founded_in	founded_by
88	The Coca-Cola Company	1886-05-08	{"Dr. John Stith Pemberton"}
89	Sony Corporation	1946-05-07	{"Masaru Ibuka", "Akio Morita"}
90	McDonald's Corporation	1955-04-15	{"Ray Kroc", "Richard McDonald", "Maurice McDonald"}
91	Intel Corporation	1968-07-18	{"Robert Noyce", "Gordon Moore"}
92	Samsung Electronics Co., Ltd.	1969-01-13	{"Lee Byung-chul"}
93	Microsoft Corporation	1975-04-04	{"Bill Gates", "Paul Allen"}
94	Apple Inc.	1976-04-01	{"Steve Jobs", "Steve Wozniak", "Ronald Wayne"}
95	Oracle Corporation	1977-06-16	{"Larry Ellison", "Bob Miner", "Ed Oates"}
96	NVIDIA Corporation	1993-04-05	{"Jensen Huang", "Chris Malachowsky", "Curtis Priem"}
97	Amazon.com Inc.	1994-07-05	{"Jeff Bezos"}
98	Google LLC	1998-09-04	{"Larry Page", "Sergey Brin"}
99	Alibaba Group Holding Limited	1999-06-28	{"Jack Ma", "18 colleagues"}
100	SAP SE	1972-04-01	{"Dietmar Hopp", "Hans-Werner Hector", "Hasso Plattner", "Klaus Tschira", "Claus Wellenreuther"}
101	LinkedIn Corporation	2002-12-28	{"Reid Hoffman", "PayPal team", "Socialnet.com team"}
102	Facebook, Inc.	2004-02-04	{"Mark Zuckerberg", "college roommates"}
103	Twitter, Inc.	2006-03-21	{"Jack Dorsey", "Biz Stone", "Evan Williams"}
104	Spotify AB	2006-04-23	{"Daniel Ek", "Martin Lorentzon"}
105	YouTube LLC	2005-02-14	{"Steve Chen", "Chad Hurley", "Jawed Karim"}
106	Tesla, Inc.	2003-07-01	{"Elon Musk", "Martin Eberhard", "Marc Tarpenning", "JB Straubel", "Ian Wright"}
107	Airbnb, Inc.	2008-08-01	{"Brian Chesky", "Joe Gebbia", "Nathan Blecharczyk"}
108	PayPal Holdings, Inc.	1998-12-01	{"Peter Thiel", "Max Levchin", "Luke Nosek", "Ken Howery"}
109	Stripe, Inc.	2010-01-01	{"Patrick Collison", "John Collison"}
110	Block, Inc.	2009-02-20	{"Jack Dorsey", "Jim McKelvey"}
111	Zoom Video Communications, Inc.	2011-04-21	{"Eric Yuan"}
112	Slack Technologies, LLC	2009-01-01	{"Stewart Butterfield", "Eric Costello", "Cal Henderson", "Serguei Mourachov"}
113	Rivian Automotive, Inc.	2009-06-23	{"RJ Scaringe"}
114	SpaceX	2002-03-14	{"Elon Musk"}
115	TikTok	2016-09-01	{"Zhang Yiming"}
116	ByteDance		{}

```
(29 rows)
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